AMC Revision

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1. Revision of inequalities: AM–GM

2000 AMC 12, Problem 12

Problem 1: Let A, M, and C be nonnegative integers such that $A + M + C = 12$. What is the maximum value of

$$A \cdot M \cdot C + A \cdot M + M \cdot C + C \cdot A?$$

2003 AMC 12A, Problem 24

Problem 2: If $a \geq b > 1$, what is the largest possible value of $\log_a(\frac{a}{b}) + \log_b(\frac{b}{a})$?

2016 AMC 12B, Problem 15

Problem 3: Label the six faces of a cube with $2, 3, 4, 5, 6, 7$, using each number once. At every vertex, multiply the labels on its three incident faces. Maximise the sum of the eight resulting products.

2020 AMC 12B, Problem 22

Problem 4: What is the maximum value of

$$\frac{(2^t - 3t)t}{4^t}$$

for real values of t?

2023 AMC 12A, Problem 23

Problem 5: How many ordered pairs of positive real numbers (a, b) satisfy the equation

$$(1 + 2a)(2 + 2b)(2a + b) = 32ab?$$

2016 AMC 12A, Problem 24

Problem 6: Among positive real parameters a, b for which $x^3 - ax^2 + bx - a$ has three real roots, choose the smallest possible a. At this minimum, b is unique. Determine b.

2. Revision of Diophantine equations

2016 AMC 10A, Problem 14

Problem 7: How many ways are there to write 2016 as the sum of twos and threes, ignoring order?

2006 AMC 10A, Problem 22

Problem 8: Two farmers value each pig at \$300 and each goat at \$210. They settle debts by transferring these animals, with animals also allowed as change. What is the smallest positive debt they can settle?

2007 AMC 10A, Problem 17

Problem 9: Suppose that m and n are positive integers such that $75m = n^3$. What is the minimum possible value of $m + n$?

2001 AMC 10, Problem 14

Problem 10: A charity earns \$2001 from 140 tickets. Each is sold either at an integer-dollar full price or at half that price. Determine the revenue from the full-price sales.

2002 AMC 10B, Problem 16

Problem 11: For how many integers n is $\dfrac{n}{20-n}$ the square of an integer?

2022 AMC 10A, Problem 17

Problem 12: Count the three-digit numbers $\overline{abc}$ with $a, b, c \in \{1, \ldots, 9\}$ satisfying

$$0.\overline{abc} = \frac{0.\overline{a} + 0.\overline{b} + 0.\overline{c}}{3}.$$

Here bars over decimal digits mean infinite repetition.

2015 AMC 10A, Problem 15

Problem 13: Consider positive fractions x/y in lowest terms. Count those whose value increases by 10% when 1 is added to both numerator and denominator.

3. Miscellaneous

2007 AMC 10A, Problem 20

Problem 14: Suppose that the number a satisfies the equation $4 = a + a^{-1}$. What is the value of $a^4 + a^{-4}$?

2001 AMC 12, Problem 12

Problem 15: How many positive integers not exceeding 2001 are multiples of 3 or 4 but not 5?

2012 AMC 10A, Problem 10

Problem 16: A circle is partitioned into 12 sectors whose angles, in degrees, are integers forming an arithmetic progression. How small can the least sector angle be?

2013 AMC 10B, Problem 17

Problem 17: Alex starts with 75 red and 75 blue tokens. A trade converts two reds into one silver and one blue; another converts three blues into one silver and one red. Trading continues until neither trade is possible. How many silver tokens remain?